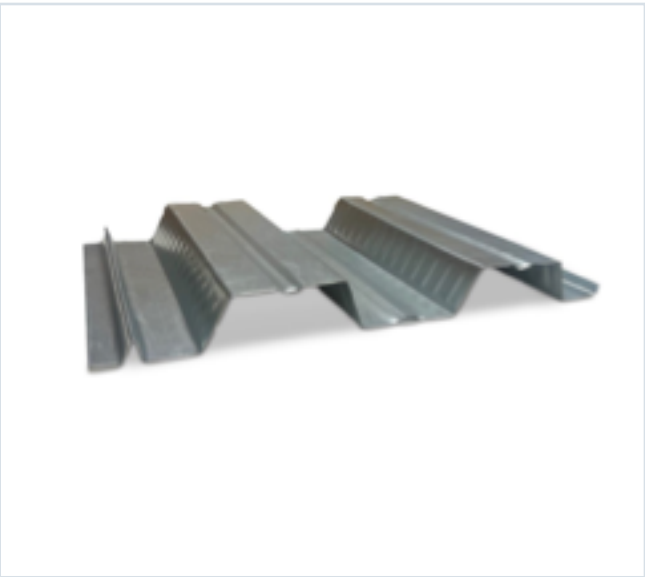


DECKING PRODUCT SUBMITTAL

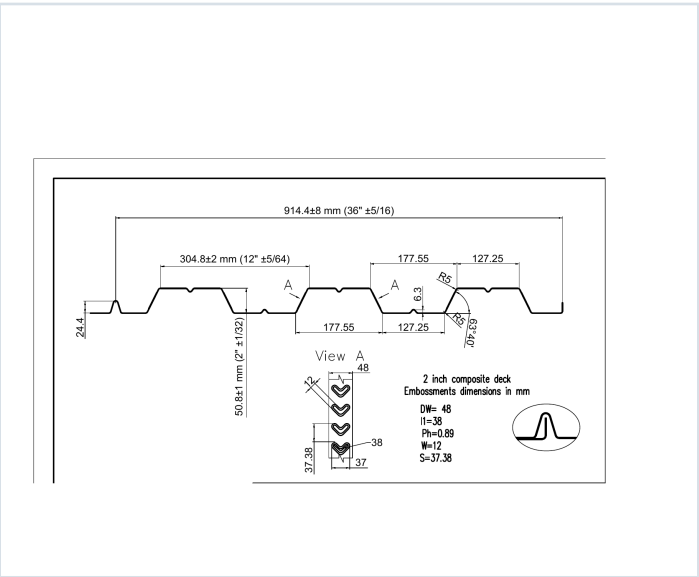
2CD-FY50-18-G40

PRODUCT	GRADE	GAUGE	COATING
2" Composite Deck	Grade 50	18 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0474	2.5	50	0.493	0.500	14760	14960	0.548	0.530

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

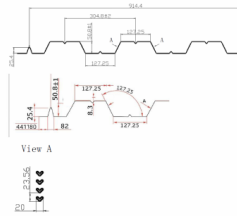
2CD-FY50-18-G40 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 54

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	E _x (ksi)	S _x (in ³)	S _y (in ³)	ASD (I _x + I _y) (in ⁴)	I _x (in ⁴)	I _y (in ⁴)
22	0.0205	16	50	0.247	0.254	2427	2427	0.31
20	0.0358	19	50	0.328	0.336	3860	3860	0.393
18	0.0474	25	50	0.485	0.500	5470	5470	0.548
16	0.0598	32	50	0.645	0.664	7921	7921	0.703

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (V _u) (kips)		Web Crippling (V _u) (kips)	
		End Bearing	Interior Bearing	End Bearing	Interior Bearing
22	388	434	476	529	579
20	278	295	327	364	398
18	365	396	433	479	519
16	463	507	557	613	664

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.

STEEL CATALOG PAGE 55

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	137	133	127	121	115	109	103	97	91	85	79
	20	181	174	166	158	150	142	134	126	118	110	102
	18	273	263	254	244	234	224	214	204	194	184	174
	16	368	355	342	329	315	301	287	273	259	245	231
Double	22	131	125	119	113	107	101	95	89	83	77	71
	20	186	178	170	162	154	146	138	130	122	114	106
	18	277	266	255	244	233	222	211	200	189	178	167
	16	367	352	337	322	307	291	275	259	243	227	211
Triple	22	176	169	162	155	148	141	134	127	120	113	106
	20	243	234	225	216	207	197	188	178	168	158	148
	18	346	334	322	310	298	286	274	262	250	238	226
	16	446	430	414	398	381	364	347	330	313	296	279

Note: 1. All section properties and ASD (I_x + I_y) uniform loads are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.
2. Loads shown in table are uniformly distributed superimposed loads in psf. Span lengths include end-to-end spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment formula used for fixed-end moments are: Single and Two-Span: $M = \frac{wL^2}{8}$; Three-Span: $M = \frac{wL^2}{16}$.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	13	13	12	11	10	9	8	7	6	5	4
	20	18	17	16	15	14	13	12	11	10	9	8
	18	27	26	25	24	23	22	21	20	19	18	17
	16	36	35	34	33	32	31	30	29	28	27	26
Double	22	13	12	11	10	9	8	7	6	5	4	3
	20	18	17	16	15	14	13	12	11	10	9	8
	18	27	26	25	24	23	22	21	20	19	18	17
	16	36	35	34	33	32	31	30	29	28	27	26
Triple	22	18	17	16	15	14	13	12	11	10	9	8
	20	24	23	22	21	20	19	18	17	16	15	14
	18	35	34	33	32	31	30	29	28	27	26	25
	16	45	44	43	42	41	40	39	38	37	36	35

Note: For loads that cause L/10 Deflection, multiply by 2.0; For loads that cause L/180 Deflection, multiply by 1.5; For loads that cause L/360 Deflection, multiply by 0.667.

PUBLISHED PERFORMANCE DATA

2CD-FY50-18-G40 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 56

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Maximum Unshored Clear Span				Maximum Unshored Clear Span			
400 (#200) 39 PSF	2x12x22 ga	7'-11"	9'-1"	9'-3"	2x12x22 ga	8'-6"	9'-9"
	2x12x20 ga	9'-3"	10'-6"	10'-10"	2x12x20 ga	10'-0"	11'-3"
	2x12x18 ga	10'-5"	12'-9"	12'-5"	2x12x18 ga	11'-2"	13'-9"
	2x12x16 ga	11'-4"	14'-2"	13'-4"	2x12x16 ga	12'-0"	14'-1"
450 (#250) 45 PSF	2x12x22 ga	7'-7"	8'-8"	8'-10"	2x12x22 ga	8'-2"	9'-5"
	2x12x20 ga	8'-10"	10'-0"	10'-4"	2x12x20 ga	9'-7"	10'-10"
	2x12x18 ga	9'-11"	12'-2"	12'-0"	2x12x18 ga	10'-9"	13'-3"
	2x12x16 ga	10'-11"	13'-8"	12'-10"	2x12x16 ga	11'-8"	14'-6"
500 (#300) 51 PSF	2x12x22 ga	7'-3"	8'-4"	8'-6"	2x12x22 ga	7'-11"	9'-1"
	2x12x20 ga	8'-6"	9'-7"	9'-11"	2x12x20 ga	9'-3"	10'-6"
	2x12x18 ga	9'-7"	11'-8"	11'-8"	2x12x18 ga	10'-5"	12'-9"
	2x12x16 ga	10'-6"	13'-3"	12'-6"	2x12x16 ga	11'-4"	14'-2"
550 (#350) 57 PSF	2x12x22 ga	6'-11"	8'-0"	8'-2"	2x12x22 ga	7'-7"	8'-9"
	2x12x20 ga	8'-3"	9'-3"	9'-6"	2x12x20 ga	8'-11"	10'-1"
	2x12x18 ga	9'-3"	11'-3"	11'-4"	2x12x18 ga	10'-0"	12'-4"
	2x12x16 ga	10'-1"	12'-9"	12'-2"	2x12x16 ga	11'-0"	13'-11"
600 (#400) 63 PSF	2x12x22 ga	6'-9"	7'-9"	7'-10"	2x12x22 ga	7'-5"	8'-6"
	2x12x20 ga	7'-11"	8'-11"	9'-2"	2x12x20 ga	8'-8"	9'-9"
	2x12x18 ga	8'-11"	10'-10"	11'-0"	2x12x18 ga	9'-9"	11'-10"
	2x12x16 ga	9'-9"	12'-4"	11'-10"	2x12x16 ga	10'-8"	13'-6"
650 (#450) 69 PSF	2x12x22 ga	6'-6"	7'-6"	7'-7"	2x12x22 ga	7'-2"	8'-3"
	2x12x20 ga	7'-9"	8'-7"	8'-10"	2x12x20 ga	8'-5"	9'-5"
	2x12x18 ga	8'-8"	10'-6"	10'-9"	2x12x18 ga	9'-5"	11'-6"
	2x12x16 ga	9'-6"	11'-11"	11'-7"	2x12x16 ga	10'-4"	12'-4"

Note
With crippling and clear span not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 61

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	375	330	290	260	230
4.5	35	400	400	400	400	395	397	384
5	39	400	400	400	400	400	370	337
5.5	44	400	400	400	400	400	400	393
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	210	180	172	157	143	131	120	110
4.5	246	239	230	191	176	160	147	138
5	304	275	250	227	208	191	175	161
5.5	354	320	291	265	242	222	204	188
6	400	367	334	304	278	255	234	216
6.5	400	400	377	344	314	288	265	244

16 ga Lightweight Concrete (115 pcf, F_c = 3,000 psi)

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	400	400	392	323	290
4.5	35	400	400	400	400	400	390	360
5	39	400	400	400	400	400	400	400
5.5	44	400	400	400	400	400	400	400
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	261	236	215	190	170	154	141	130
4.5	317	287	261	236	216	200	184	170
5	377	342	311	284	260	238	219	202
5.5	436	398	362	330	303	278	254	236
6	400	400	400	380	348	319	294	272
6.5	400	400	400	400	393	361	333	307